

Converting X-ray Linear Polarization to Circular Polarization Using a Crystal Wave Plate

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Abstract

Converting a linearly polarized X-ray beam into a circularly polarized beam is a standard technique at synchrotron radiation facilities. The key optical element is a **quarter-wave plate (QWP)**, a birefringent crystal that introduces a 90° ($\lambda/4$) phase retardation between the two orthogonal components of the electric field.

This document presents a self-contained theoretical treatment covering: (i) the Jones and Stokes formalisms and their connection via the Pauli matrices; (ii) the Jones and Mueller matrices of a general retarder at an arbitrary orientation angle χ ; (iii) the particular case of the QWP converting horizontal linear polarization into right circular polarization; (iv) the experimental Stokes polarimetry protocol used at synchrotron beamlines; and (v) the general crystal phase plate with complex amplitudes r_s , r_p and the resulting degree of circular polarization as a function of χ and the relative phase retardation $\Delta\phi$.

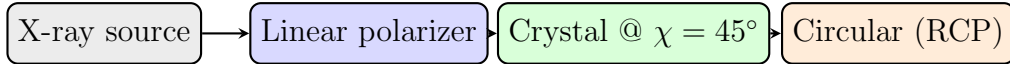
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1 Introduction

Synchrotron X-ray beams produced by bending magnets or undulators are naturally linearly polarized in the horizontal plane of the storage ring. Many X-ray spectroscopy and scattering experiments — magnetic circular dichroism (MCD), resonant magnetic X-ray scattering [1, 2], chiral diffraction — require circularly polarized beams. The standard approach is to insert a **crystal phase plate** in the beam, which acts as the X-ray analogue of a quarter-wave plate [3–5].

To convert horizontal linear polarization to circular polarization, the crystal must be oriented with its birefringence axis at $\chi = 45^\circ$ to the incoming polarization direction, so that both σ - and π -polarization components are equally excited and acquire a relative phase shift of $\pi/2$. The optical layout is:



The optical properties of perfect crystals offer a powerful mechanism for manipulating polarization through birefringence near a Bragg reflection. The σ - and π -polarization components experience different complex reflectivities r_s and r_p , whose relative magnitude and phase can be tuned by rocking the crystal angle. The phase retardation $\Delta\phi = \arg(r_s) - \arg(r_p)$ reaches $\pm\pi/2$ at a detuning of approximately half the Darwin width from the Bragg peak, providing the ideal quarter-wave condition.

Practically, thin (a few hundred microns) perfect crystals (silicon or diamond) are used as phase plates at hard X-ray beamlines (ESRF, APS, SPring-8) because they can be tuned to high near-equal transmission, phase shift close to $\pi/2$, and offer excellent mechanical stability [4, 5].

2 Jones and Mueller–Stokes Calculus Treatment

2.1 Jones Vector and Stokes Parameters

We start from the physical, real-valued electric field of a fully coherent monochromatic plane wave propagating along $\hat{z}$, written exactly as in Eq. (1) of Ref. [3]:

$$\mathcal{E}(t, \mathbf{r}) = \text{Re}[(E_x \hat{x} + E_y \hat{y}) e^{-i(\omega t - kz)}] \quad (1)$$

where the coordinate system is right-handed, $\hat{x} \times \hat{y} = \hat{z} = \hat{k}$, and the complex amplitudes carry the absolute phases, $E_{x,y} = |E_{x,y}| e^{i\phi_{x,y}}$. Writing $\tau = \omega t - kz$ for the propagator, the real field components are

$$\mathcal{E}_x(t) = |E_x| \cos(\tau - \phi_x), \quad \mathcal{E}_y(t) = |E_y| \cos(\tau - \phi_y) \quad (2)$$

which reproduces Eq. (2) of Ref. [3] with $V_1 = E_x$, $V_2 = E_y$, $\hat{e}_1 = \hat{x}$, $\hat{e}_2 = \hat{y}$.

The choice of the time factor $e^{-i\omega t}$ in Eq. (1) is a genuine convention (the complex conjugate representation, $e^{+i\omega t}$, is equally common in the literature and describes the same physical field). It must be stated explicitly because every statement about the *sign* of the circular polarization — and therefore about handedness — ultimately traces back to it: conjugating the representation flips the sign of S_3 defined below while leaving S_0 , S_1 , S_2 unchanged.

The **Jones vector** [6] is the complex envelope of Eq. (1), i.e. the pair of complex amplitudes:

$$\mathbf{E} = \begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} |E_x| e^{i\phi_x} \\ |E_y| e^{i\phi_y} \end{pmatrix} \quad (3)$$

where E_x and E_y are the complex amplitudes of the horizontal ($\hat{x}$) and vertical ($\hat{y}$) field components, and ϕ_x, ϕ_y are their absolute phases.

The four real, observable **Stokes parameters** [7, 8] are defined as:

$$S_0 = |E_x|^2 + |E_y|^2 \quad (4)$$

$$S_1 = |E_x|^2 - |E_y|^2 \quad (5)$$

$$S_2 = 2 \operatorname{Re}(E_x E_y^*) \quad (6)$$

$$S_3 = 2 \operatorname{Im}(E_x E_y^*) \quad (7)$$

Sign convention for S_3 . The sign of S_3 is a frequent source of confusion because both $2 \operatorname{Im}(E_x^* E_y)$ and $2 \operatorname{Im}(E_x E_y^*)$ appear in the literature, differing by an overall sign. Here we adopt $S_3 = 2 \operatorname{Im}(E_x^* E_y)$ because it is the unique choice consistent with the Pauli trace projection $S_k = \operatorname{Tr}(\boldsymbol{\sigma}_k \boldsymbol{\rho})$ of Eq. (10) and the standard Pauli matrix $\boldsymbol{\sigma}_3$ of Eq. (8), on which all Mueller matrices in this document are built. With this choice, and with the time convention of Eq. (1), the state $\mathbf{E} = (1, i)^T / \sqrt{2}$ has $S_3 = +1$: its field vector rotates counter-clockwise when looking toward the source, i.e. it is *left* circular polarization (LCP) in the classical optics naming, exactly as in Ref. [3]. Conversely $\mathbf{E} = (1, -i)^T / \sqrt{2}$ has $S_3 = -1$ and is right circular polarization (RCP).

S_0 is the total intensity, S_1 measures linear horizontal/vertical polarization preference, S_2 measures $\pm 45^\circ$ linear polarization, and S_3 measures circular polarization content.

Parameter	Measures	Extreme values (normalized)	
S_0	Total intensity	$S_0 > 0$ always	conserved through lossless optics
S_1	H vs. V linear	+1: horiz. (H)	-1: vertical (V)
S_2	$\pm 45^\circ$ linear	+1: $+45^\circ$	-1: -45°
S_3	Circular content	+1: LCP	-1: RCP

Table 1: Physical interpretation of the Stokes parameters (normalized, $S_0 = 1$).

Pauli Matrix Representation

The four **Pauli matrices** (including the 2×2 identity) form a complete orthogonal basis for all 2×2 Hermitian matrices:

$$\boldsymbol{\sigma}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \boldsymbol{\sigma}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \boldsymbol{\sigma}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\sigma}_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (8)$$

They satisfy the orthogonality relation $\operatorname{Tr}(\boldsymbol{\sigma}_m \boldsymbol{\sigma}_n) = 2\delta_{mn}$ and the product rule $\boldsymbol{\sigma}_j \boldsymbol{\sigma}_k = \delta_{jk} \boldsymbol{\sigma}_0 + i\varepsilon_{jkl} \boldsymbol{\sigma}_l$, where ε_{jkl} is the Levi-Civita symbol.

The **density matrix** $\boldsymbol{\rho} = \mathbf{E} \mathbf{E}^\dagger$ (called the *coherency matrix* in classical optics [7]) provides a 2×2 Hermitian representation of the polarization state:

$$\boldsymbol{\rho} = \begin{pmatrix} \langle E_x E_x^* \rangle & \langle E_x E_y^* \rangle \\ \langle E_y E_x^* \rangle & \langle E_y E_y^* \rangle \end{pmatrix} = \frac{1}{2} \sum_{k=0}^3 S_k \boldsymbol{\sigma}_k \quad (9)$$

The Stokes parameters are then the projections of $\boldsymbol{\rho}$ onto the Pauli basis via the trace inner product:

$$S_k = \text{Tr}(\boldsymbol{\sigma}_k \boldsymbol{\rho}), \quad k = 0, 1, 2, 3 \quad (10)$$

This compact formula reproduces Eqs. (4)–(7) exactly and is the foundation of the Jones-to-Mueller conversion used throughout this document (see Section 2.3). Conversely, expanding the sum $\boldsymbol{\rho} = \frac{1}{2} \sum_k S_k \boldsymbol{\sigma}_k$ gives the density matrix explicitly in terms of the Stokes parameters:

$$\boldsymbol{\rho} = \frac{1}{2} \begin{pmatrix} S_0 + S_1 & S_2 - iS_3 \\ S_2 + iS_3 & S_0 - S_1 \end{pmatrix} \quad (11)$$

a form that provides a quick consistency check of all sign conventions: the imaginary part of the off-diagonal element carries S_3 , with $\rho_{21} = \langle E_y E_x^* \rangle = \frac{1}{2}(S_2 + iS_3)$.

Conventions in the literature. The linked set of sign choices made above — time factor, Stokes definitions, Pauli representation, handedness naming — is by no means universal, and the resulting ambiguities are a well-documented source of errors. Both $S_3 = 2\text{Im}(E_x^* E_y)$ and its negative appear in standard references, since conjugating the analytic-signal representation ($e^{-i\omega t} \leftrightarrow e^{+i\omega t}$) describes the same physical field while flipping the sign of S_3 . Handedness *naming* adds an independent choice: classical optics names the rotation sense as seen by an observer looking toward the source [7], while the IEEE/antenna and particle-physics helicity conventions look along the propagation direction and obtain the opposite name; Jackson [9] explicitly warns that his “positive helicity” corresponds to *left* circular polarization in the traditional optics nomenclature. The representation freedom in the Pauli-matrix assignment was analyzed early on by McMaster [10], and in radio astronomy the confusion was severe enough that the IAU fixed the sign of Stokes V by resolution, whose reconciliation with the IEEE definitions required a dedicated paper [11]. In the X-ray magnetic scattering literature, the Blume–Gibbs density-matrix formulation [1] orders the polarization components differently again. For these reasons no single formula can be quoted safely without its companions; in this document the complete chain is fixed once — Eq. (1), Eqs. (4)–(7), Eq. (8), and the anchor state $(1, i)^T / \sqrt{2} \mapsto S_3 = +1$ (LCP) — matching Ref. [3], and everything else is derived from it.

2.2 Degree of Polarization and Poincaré Sphere

The **degree of polarization** (DOP) quantifies how far a beam departs from complete unpolarized light:

$$\text{DOP} = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0} \in [0, 1] \quad (12)$$

DOP = 1 for fully coherent (polarized) light; DOP = 0 for completely unpolarized light. The degree of linear and circular polarization are:

$$\text{DOLP} = \frac{\sqrt{S_1^2 + S_2^2}}{S_0}, \quad \text{DOCP} = \frac{S_3}{S_0} \in [-1, +1] \quad (13)$$

The **Poincaré sphere** provides a geometric representation: a fully polarized state corresponds to a point on the sphere of radius S_0 at position (S_1, S_2, S_3) .

- **North pole** $(0, 0, +S_3)$: left circular polarization (LCP)

- **South pole** $(0, 0, -S_3)$: right circular polarization (RCP)
- **Equator** ($S_3 = 0$): all linear polarization states
- $+S_1$ axis: horizontal linear; $-S_1$ axis: vertical linear
- $+S_2$ axis: $+45^\circ$ linear; $-S_2$ axis: -45° linear

Optical elements act as rotations on the Poincaré sphere (for lossless retarders) or as contractions toward the origin (for partial polarizers).

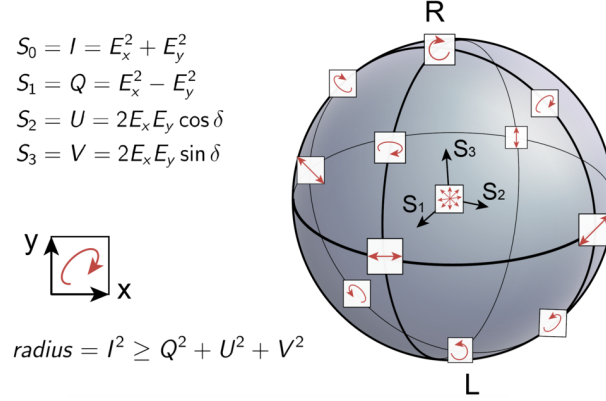


Figure 1: The Poincaré sphere. <https://commons.wikimedia.org/w/index.php?curid=90694298>
Note that the R/L handedness labels in this figure follow the opposite naming convention to the one adopted here (Section 2.1), where $S_3 = +1$ is LCP.

2.3 Jones Matrices and Mueller Matrices

An optical element transforms the polarization state of a beam. In the two formalisms:

- In the **Jones formalism**, a 2×2 complex matrix $\mathbf{J}$ transforms the Jones vector: $\mathbf{E}' = \mathbf{J} \mathbf{E}$. This is valid for fully coherent beams only.
- In the **Mueller–Stokes formalism** [12, 13], a real 4×4 matrix $\mathbf{M}$ transforms the Stokes vector: $\mathbf{S}' = \mathbf{M} \mathbf{S}$. This is valid for any degree of coherence and polarization.

For a fully coherent beam, the two formalisms are equivalent. The Mueller matrix corresponding to a Jones matrix $\mathbf{J}$ is obtained via the Pauli trace formula. Since the density matrix transforms as $\boldsymbol{\rho}' = \mathbf{J} \boldsymbol{\rho} \mathbf{J}^\dagger$, projecting onto the Pauli basis (Eq. 10) gives each Mueller matrix element as:

$$M_{mn} = \frac{1}{2} \text{Tr}(\boldsymbol{\sigma}_m \mathbf{J} \boldsymbol{\sigma}_n \mathbf{J}^\dagger), \quad m, n \in \{0, 1, 2, 3\} \quad (14)$$

When several optical elements are placed in series, their Jones matrices (or Mueller matrices) are multiplied in order from right to left:

$$\mathbf{J}_{\text{total}} = \mathbf{J}_N \cdots \mathbf{J}_2 \mathbf{J}_1, \quad \mathbf{M}_{\text{total}} = \mathbf{M}_N \cdots \mathbf{M}_2 \mathbf{M}_1 \quad (15)$$

2.4 Rotation of an Optical Element by Angle χ

When an optical element is physically rotated by angle χ about the beam axis, its Jones matrix in the lab frame becomes:

$$\mathbf{J}^{(\chi)} = \mathbf{R}(-\chi) \cdot \mathbf{J}^{(0)} \cdot \mathbf{R}(\chi) \quad (16)$$

where the Jones rotation matrix is:

$$\mathbf{R}(\chi) = \begin{pmatrix} \cos \chi & \sin \chi \\ -\sin \chi & \cos \chi \end{pmatrix} \quad (17)$$

Derivation of Eq. (16) and convention note

Eq. (16) follows from a transparent three-step argument. With the rotation matrix $\mathbf{R}(\chi)$ as defined in Eq. (17):

1. **Align.** The component's axes are rotated by $+\chi$ relative to the lab. Project the lab input field onto the component's own frame:

$$\mathbf{E}_{\text{in,comp}} = \mathbf{R}(\chi) \mathbf{E}_{\text{in,lab}} \quad (18)$$

($\mathbf{R}(\chi)$ maps lab coordinates to component coordinates because $\mathbf{R}(\chi)^{-1} = \mathbf{R}(-\chi)$, so the inverse transform from component to lab is $\mathbf{R}(-\chi)$.)

2. **Act.** Apply the component's intrinsic Jones matrix in its own frame:

$$\mathbf{E}_{\text{out,comp}} = \mathbf{J}^{(0)} \mathbf{E}_{\text{in,comp}} \quad (19)$$

3. **Realign.** Rotate back to lab coordinates with $\mathbf{R}(-\chi)$:

$$\mathbf{E}_{\text{out,lab}} = \mathbf{R}(-\chi) \mathbf{E}_{\text{out,comp}} \quad (20)$$

Combining gives $\mathbf{E}_{\text{out,lab}} = \mathbf{R}(-\chi) \mathbf{J}^{(0)} \mathbf{R}(\chi) \mathbf{E}_{\text{in,lab}}$, which identifies $\mathbf{J}^{(\chi)}$ as in Eq. (16).

Quick verification. A horizontal polarizer ($\mathbf{J}^{(0)} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$) rotated by $\chi = 90^\circ$ should become a vertical polarizer. Using Eq. (16):

$$\mathbf{J}^{(90^\circ)} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad \checkmark \quad (21)$$

Convention note. $\mathbf{R}(\chi)$ as defined in Eq. (17) is the *passive* rotation matrix: it expresses the components of a fixed field in a coordinate frame rotated by $+\chi$ (from $+\hat{x}$ toward $+\hat{y}$). It is the transpose (inverse) of the standard counter-clockwise *active* rotation matrix $\begin{bmatrix} \cos \chi & -\sin \chi \\ \sin \chi & \cos \chi \end{bmatrix}$ used in some optics references. With the active matrix the sandwich for a rotated element reads $\mathbf{R}_{\text{act}}(\chi) \mathbf{J}^{(0)} \mathbf{R}_{\text{act}}(-\chi)$ — identical physics, opposite letter ordering. In either case, positive χ in Eq. (16) means that the element's fast axis is rotated from $+\hat{x}$ toward $+\hat{y}$; this is the same sense as the polarizer angle θ in Section 3, whose transmission axis is $(\cos \theta, \sin \theta)$. Reversing the sign convention of the sandwich would flip the sign of $\sin 2\chi$ (equivalently of S_2 and S_3 generated from horizontal input) everywhere below. When importing Jones matrices from other sources always check which

sign convention $\mathbf{R}(\chi)$ uses. Note also that Ref. [3] writes its frame-change matrix $R(\eta)$ (just above its Eqs. (24)–(25)) with the opposite sign of the off-diagonal terms, while its Eqs. (24)–(25) for the Stokes parameters correspond exactly to the Mueller rotation matrix $\mathbf{R}_M(\chi)$ of Eq. (22) below, i.e. to the component transformation of Eq. (17).

The analogous Mueller rotation matrix is:

$$\mathbf{R}_M(\chi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\chi & \sin 2\chi & 0 \\ 0 & -\sin 2\chi & \cos 2\chi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (22)$$

$\mathbf{R}_M(\chi)$ plays the same role for Stokes vectors that $\mathbf{R}(\chi)$ plays for Jones vectors: $\mathbf{S}' = \mathbf{R}_M(\chi) \mathbf{S}$ gives the Stokes parameters in a frame rotated by $+\chi$. The key property is that a *physical* rotation by χ appears as a rotation by 2χ in Stokes space, because the Stokes parameters are quadratic in the field amplitudes. Note that S_0 (total intensity) and S_3 (circular content) are invariant under rotation, while (S_1, S_2) rotate as a two-dimensional vector.

The rotated Mueller matrix is:

$$\mathbf{M}^{(\chi)} = \mathbf{R}_M(-\chi) \cdot \mathbf{M}^{(0)} \cdot \mathbf{R}_M(\chi) \quad (23)$$

2.5 Jones and Mueller Matrices of a General Retarder

A **linear retarder** (wave plate) introduces a phase retardation δ between its fast and slow eigenpolarization axes. With the fast axis aligned along $\hat{x}$ ($\chi = 0$), the Jones matrix is:

$$\mathbf{J}_{\text{ret}}^{(0)} = e^{-i\delta/2} \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix} \quad (24)$$

The global phase factor $e^{-i\delta/2}$ is physically irrelevant for intensity measurements and is often dropped. Applying the Pauli trace formula (Eq. 14) to $\mathbf{J}_{\text{ret}}^{(0)}$, the corresponding aligned Mueller matrix of the general retarder is:

$$\mathbf{M}_{\text{ret}}^{(0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \delta & -\sin \delta \\ 0 & 0 & \sin \delta & \cos \delta \end{pmatrix} \quad (25)$$

i.e. a rotation by the angle δ in the (S_2, S_3) plane of the Poincaré sphere, leaving S_0 and S_1 untouched. For a **quarter-wave plate** (QWP), $\delta = \pi/2$:

$$\mathbf{J}_{\text{QWP}}^{(0)} = e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad (26)$$

Setting $\delta = \pi/2$ in Eq. (25) (equivalently, using Eq. 14), the corresponding aligned Mueller matrix is:

$$\mathbf{M}_{\text{QWP}}^{(0)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix} \quad (27)$$

This matrix leaves S_0 and S_1 unchanged (H/V polarization unaffected since the fast axis is along $\hat{x}$) and maps $S_2 \rightarrow -S_3$, $S_3 \rightarrow +S_2$: it converts $\pm 45^\circ$ linear polarization into circular polarization.

For a general retarder with fast axis rotated to angle χ , applying Eq. (23) to $\mathbf{M}_{\text{ret}}^{(0)}$ gives the full Mueller matrix:

$$\mathbf{M}_{\text{ret}}^{(\chi, \delta)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos^2 2\chi + \sin^2 2\chi \cos \delta & \sin 2\chi \cos 2\chi (1 - \cos \delta) & + \sin 2\chi \sin \delta \\ 0 & \sin 2\chi \cos 2\chi (1 - \cos \delta) & \sin^2 2\chi + \cos^2 2\chi \cos \delta & - \cos 2\chi \sin \delta \\ 0 & - \sin 2\chi \sin \delta & + \cos 2\chi \sin \delta & \cos \delta \end{pmatrix} \quad (28)$$

One verifies immediately: $\delta = \pi/2$, $\chi = 0$ recovers Eq. (27) ✓.

2.6 QWP at $\chi = 45^\circ$: Jones and Mueller Matrices

Applying the rotation sandwich (Eq. 16) to $\mathbf{J}_{\text{QWP}}^{(0)}$ with $\chi = 45^\circ$:

$$\mathbf{J}_{\text{QWP}}^{(45^\circ)} = \mathbf{R}(-45^\circ) \cdot e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \cdot \mathbf{R}(45^\circ) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \quad (29)$$

Setting $\delta = \pi/2$ and $\chi = 45^\circ$ in Eq. (28):

$$\mathbf{M}_{\text{QWP}}^{(45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \quad (30)$$

This matrix swaps $S_1 \leftrightarrow -S_3$ while leaving S_0 and S_2 unchanged. It converts horizontal linear polarization ($S_1 = 1, S_3 = 0$) into right circular polarization ($S_1 = 0, S_3 = -1$).

Equivalently, applying $\mathbf{R}_M(-45^\circ) \mathbf{M}_{\text{QWP}}^{(0)} \mathbf{R}_M(45^\circ)$ with $\mathbf{R}_M(\pm 45^\circ)$ evaluated at $2\chi = \pm 90^\circ$ yields the same result as Eq. (30). ✓

2.7 Input Beam: Horizontal Linear Polarization

The Jones vector of a horizontally polarized beam (normalized, $E_0 = 1$) is:

$$\mathbf{E}_{\text{in}} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (31)$$

Applying Eqs. (4)–(7):

$$\mathbf{S}_{\text{in}} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad (32)$$

This state lies on the equator of the Poincaré sphere at the $+S_1$ pole.

2.8 Output Beam: Right Circular Polarization

2.8.1 Jones route

Applying Eq. (29) to the input Jones vector:

$$\mathbf{E}_{\text{out}} = \mathbf{J}_{\text{QWP}}^{(45^\circ)} \mathbf{E}_{\text{in}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (33)$$

Hence:

$$\mathbf{E}_{\text{out}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \quad (34)$$

This is the canonical Jones vector for right circular polarization. Computing the Stokes parameters from Eqs. (4)–(7):

$$S_0 = \left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{-i}{\sqrt{2}} \right|^2 = \frac{1}{2} + \frac{1}{2} = 1 \quad (35)$$

$$S_1 = \left| \frac{1}{\sqrt{2}} \right|^2 - \left| \frac{-i}{\sqrt{2}} \right|^2 = \frac{1}{2} - \frac{1}{2} = 0 \quad (36)$$

$$S_2 = 2 \operatorname{Re} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} \right) = 2 \operatorname{Re} \left(\frac{-i}{2} \right) = 0 \quad (37)$$

$$S_3 = 2 \operatorname{Im} \left(\frac{1}{\sqrt{2}} \cdot \frac{-i}{\sqrt{2}} \right) = 2 \operatorname{Im} \left(\frac{-i}{2} \right) = -1 \quad (38)$$

The corresponding density matrix, from $\boldsymbol{\rho} = \mathbf{E}_{\text{out}} \mathbf{E}_{\text{out}}^\dagger$ or equivalently from Eq. (11) with $\mathbf{S} = (1, 0, 0, -1)^T$, is:

$$\boldsymbol{\rho}_{\text{RCP}} = \frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \quad (39)$$

2.8.2 Mueller route

Applying $\mathbf{M}_{\text{QWP}}^{(45^\circ)}$ (Eq. 30) to $\mathbf{S}_{\text{in}}$ (Eq. 32):

$$\mathbf{S}_{\text{out}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (40)$$

The Mueller route reproduces exactly the Stokes vector obtained from the Jones route: both give $S_3 = -1$, i.e. right circular polarization in the convention of Section 2.1. This agreement is automatic because Eq. (7) and the Pauli trace formula (Eq. 14) use the same sign convention.

2.8.3 Time-domain picture

In real space and time, inserting $\mathbf{E}_{\text{out}}$ of Eq. (34) into Eqs. (1)–(2) (at $z = 0$, with $\phi_x = 0$, $\phi_y = -\pi/2$), the output field components are:

$$E_x(t) = \frac{1}{\sqrt{2}} \cos \omega t, \quad E_y(t) = -\frac{1}{\sqrt{2}} \sin \omega t \quad (41)$$

The tip of the electric field vector traces a **circle** of radius $1/\sqrt{2}$ in the (x, y) plane, confirming circular polarization. The rotation is clockwise when viewed looking toward the source — right circular polarization (RCP) in the classical optics naming adopted here (Section 2.1 and Ref. [3]).

2.8.4 Summary: both routes give RCP

$$\mathbf{S}_{\text{out}} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (\text{RCP}), \quad \text{DOCP} = -1 \quad (42)$$

The state sits at the **south pole** of the Poincaré sphere.

2.9 Left Circular Polarization: QWP at $\chi = -45^\circ$

Rotating the QWP to $\chi = -45^\circ$ reverses the sign of the off-diagonal terms in the Jones matrix:

$$\mathbf{J}_{\text{QWP}}^{(-45^\circ)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \implies \mathbf{E}_{\text{out}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad \mathbf{S}_{\text{out}} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ +1 \end{pmatrix} \quad (\text{LCP}) \quad (43)$$

The corresponding Mueller matrix is:

$$\mathbf{M}_{\text{QWP}}^{(-45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (44)$$

which maps horizontal linear ($S_1 = 1$) to $S_3 = +1$ (LCP), as expected. The density matrix of the LCP state, from $\boldsymbol{\rho} = \mathbf{E}_{\text{out}} \mathbf{E}_{\text{out}}^\dagger$ or from Eq. (11) with $\mathbf{S} = (1, 0, 0, +1)^T$, is:

$$\boldsymbol{\rho}_{\text{LCP}} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix} \quad (45)$$

Note that $\boldsymbol{\rho}_{\text{RCP}}$ and $\boldsymbol{\rho}_{\text{LCP}}$ differ only in the sign of the imaginary off-diagonal element, which carries S_3 (Eq. 11).

2.10 Summary of Key Polarization States and Optical Elements

Table 2 collects the three equivalent representations — Jones vector, Stokes vector, and density matrix — of the basic fully polarized states used throughout this document, in the conventions of Section 2.1.

State	Jones $\mathbf{E}$	Stokes $\mathbf{S}^T$	Density matrix $\boldsymbol{\rho}$
Linear horizontal (0°)	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$(1, 1, 0, 0)$	$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$
Linear $+45^\circ$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$(1, 0, 1, 0)$	$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$
Linear vertical (90°)	$\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	$(1, -1, 0, 0)$	$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$
Right circular (RCP)	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$	$(1, 0, 0, -1)$	$\frac{1}{2} \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$
Left circular (LCP)	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$	$(1, 0, 0, +1)$	$\frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}$

Table 2: Basic fully polarized states (normalized, $S_0 = 1$) in the three equivalent representations, using the conventions of Section 2.1 ($e^{-i(\omega t - kz)}$ time factor; $S_3 = 2 \text{Im}(E_x^* E_y)$; handedness named looking toward the source). Jones vectors are defined up to an arbitrary global phase. The density matrices follow from $\boldsymbol{\rho} = \mathbf{E}\mathbf{E}^\dagger$ or equivalently from Eq. (11).

2.11 Summary of Key Jones and Mueller Matrices

Element	Jones matrix $\mathbf{J}$	Mueller matrix $\mathbf{M}$
QWP, fast axis at $\chi = 0^\circ$	$e^{-i\pi/4} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & +1 & 0 \end{pmatrix}$
QWP, fast axis at $\chi = +45^\circ$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$
QWP, fast axis at $\chi = -45^\circ$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & +i \\ +i & 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$
General retarder		
fast axis at χ , retardation δ	$e^{-i\delta/2} \mathbf{R}(-\chi) \begin{pmatrix} 1 & 0 \\ 0 & e^{i\delta} \end{pmatrix} \mathbf{R}(\chi)$ see Eq. (28)	
Rotation by χ	$\begin{pmatrix} \cos \chi & \sin \chi \\ -\sin \chi & \cos \chi \end{pmatrix}$	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\chi & \sin 2\chi & 0 \\ 0 & -\sin 2\chi & \cos 2\chi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

Table 3: Jones and Mueller matrices for key optical elements. The general retarder entry uses the rotation sandwich $\mathbf{R}(-\chi) \mathbf{J}^{(0)} \mathbf{R}(\chi)$ of Eq. (16) (see Section 2.4); the QWP entries include the $e^{-i\delta/2} = e^{-i\pi/4}$ prefactor of Eq. (24), which is why no residual global phase appears at $\chi = \pm 45^\circ$. The doubling $\chi \rightarrow 2\chi$ in the Mueller rotation matrix reflects the quadratic dependence of Stokes parameters on the field amplitudes.

3 Experimental Polarization Analysis

3.1 Principle

A linear polarizer placed in the beam transmits only the projection of the electric field onto its transmission axis. By recording the transmitted intensity $I(\theta)$ as a function of the polarizer rotation angle θ , and by optionally inserting a quarter-wave plate (QWP) analyzer before the polarizer, all four Stokes parameters of an *arbitrary* polarization state (S_0, S_1, S_2, S_3) can be determined.

3.2 Transmitted Intensity Formula

For a linear polarizer at angle θ to the horizontal, the Jones matrix is:

$$\mathbf{J}_{LP}(\theta) = \begin{pmatrix} \cos^2 \theta & \cos \theta \sin \theta \\ \cos \theta \sin \theta & \sin^2 \theta \end{pmatrix} \quad (46)$$

The Mueller matrix of a linear polarizer at angle θ is:

$$\mathbf{M}_{LP}(\theta) = \frac{1}{2} \begin{pmatrix} 1 & \cos 2\theta & \sin 2\theta & 0 \\ \cos 2\theta & \cos^2 2\theta & \sin 2\theta \cos 2\theta & 0 \\ \sin 2\theta & \sin 2\theta \cos 2\theta & \sin^2 2\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad (47)$$

Applying this to an arbitrary incident Stokes vector $\mathbf{S} = (S_0, S_1, S_2, S_3)^T$, the detected intensity is the first component of $\mathbf{M}_{LP}(\theta) \mathbf{S}$:

$$I(\theta) = \frac{1}{2} (S_0 + S_1 \cos 2\theta + S_2 \sin 2\theta) \quad (48)$$

This is a **sinusoidal function of 2θ** with mean $S_0/2$, cosine amplitude $S_1/2$, and sine amplitude $S_2/2$. Note that S_3 does *not* appear: a linear polarizer alone is insensitive to circular polarization.

3.3 Measurements at Key Angles: General Case

Four strategically chosen orientations of the polarizer are sufficient to determine S_0 , S_1 , and S_2 from Eq. (48).

$\theta = 0^\circ$ (**Horizontal transmission axis**)

$$I_H \equiv I(0^\circ) = \frac{1}{2} (S_0 + S_1) \quad (49)$$

$\theta = 90^\circ$ (**Vertical transmission axis**)

$$I_V \equiv I(90^\circ) = \frac{1}{2} (S_0 - S_1) \quad (50)$$

$\theta = 45^\circ$ (**+45° transmission axis**)

$$I_{45} \equiv I(45^\circ) = \frac{1}{2} (S_0 + S_2) \quad (51)$$

$\theta = 135^\circ$ (-45° **transmission axis**)

$$I_{135} \equiv I(135^\circ) = \frac{1}{2}(S_0 - S_2) \quad (52)$$

From these four measurements the first three Stokes parameters are recovered by simple linear combinations:

$$S_0 = I_H + I_V = I_{45} + I_{135} \quad (53)$$

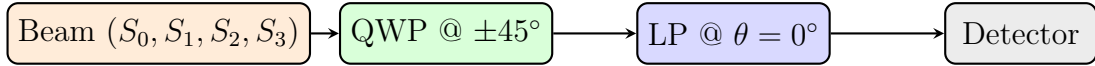
$$S_1 = I_H - I_V \quad (54)$$

$$S_2 = I_{45} - I_{135} \quad (55)$$

Equations (53)–(55) are valid for *any* incident polarization state. They show that S_1 is the contrast between horizontal and vertical transmissions, and S_2 is the contrast between the two diagonal transmissions.

3.4 Measuring S_3 : Adding a QWP Analyzer

Since a linear polarizer alone cannot detect S_3 , a QWP must be inserted immediately before the polarizer. Its role is to convert circular polarization into linear polarization that the polarizer can then analyse.



Applying $\mathbf{M}_{\text{QWP}}^{(\pm 45^\circ)}$ (Eq. 30) to the general Stokes vector, then transmitting through LP at $\theta = 0^\circ$:

$$I_+ \equiv I[\text{QWP}(+45^\circ), \text{LP}(0^\circ)] = \frac{1}{2}(S_0 + S_3) \quad (56)$$

$$I_- \equiv I[\text{QWP}(-45^\circ), \text{LP}(0^\circ)] = \frac{1}{2}(S_0 - S_3) \quad (57)$$

The general relation for S_3 is:

$$S_3 = I_+ - I_- = \frac{I_{\text{QWP}(+45^\circ)} - I_{\text{QWP}(-45^\circ)}}{1} \quad (58)$$

or equivalently, normalized to the total intensity:

$$\text{DOCP} = \frac{S_3}{S_0} = \frac{I_+ - I_-}{I_+ + I_-} \quad (59)$$

Configuration	Detected intensity	Stokes relation
LP at $\theta = 0^\circ$	$I_H = \frac{1}{2}(S_0 + S_1)$	$S_1 = 2I_H - S_0$
LP at $\theta = 90^\circ$	$I_V = \frac{1}{2}(S_0 - S_1)$	$S_0 = I_H + I_V$
LP at $\theta = 45^\circ$	$I_{45} = \frac{1}{2}(S_0 + S_2)$	$S_2 = 2I_{45} - S_0$
LP at $\theta = 135^\circ$	$I_{135} = \frac{1}{2}(S_0 - S_2)$	check: $S_0 = I_{45} + I_{135}$
QWP(+45°)+LP(0°)	$I_+ = \frac{1}{2}(S_0 + S_3)$	$S_3 = I_+ - I_-$
QWP(-45°)+LP(0°)	$I_- = \frac{1}{2}(S_0 - S_3)$	$\text{DOCP} = (I_+ - I_-)/(I_+ + I_-)$

Table 4: General six-measurement Stokes polarimetry protocol, valid for any incident polarization state (S_0, S_1, S_2, S_3) .

3.5 Particular Case: Circular Polarization

For the RCP beam produced by the crystal QWP at $\chi = +45^\circ$ (Section 2.8), the Stokes vector is $\mathbf{S} = I_0(1, 0, 0, -1)^T$, i.e. $S_1 = S_2 = 0$ and $S_3 = -S_0 = -I_0$. Substituting into the general formulas:

Linear polarizer scan.

$$I(\theta) = \frac{1}{2}(I_0 + 0 \cdot \cos 2\theta + 0 \cdot \sin 2\theta) = \frac{I_0}{2} \quad (60)$$

$$I(\theta) = \frac{I_0}{2} \quad \forall \theta \quad (61)$$

The intensity is **constant regardless of polarizer angle** — the definitive fingerprint of a fully circularly polarized beam. Because $S_1 = S_2 = 0$, the sinusoidal modulation of Eq. (48) vanishes completely.

Key angle measurements. From Eqs. (49)–(52):

$$I_H = I_V = I_{45} = I_{135} = \frac{I_0}{2} \quad (62)$$

All four linear polarizer positions give the same reading, confirming $S_1 = S_2 = 0$.

QWP analyzer for S_3 . From Eqs. (56)–(57):

$$I_+ = \frac{1}{2}(I_0 - I_0) = 0 \quad (\text{minimum}) \quad (63)$$

$$I_- = \frac{1}{2}(I_0 + I_0) = I_0 \quad (\text{maximum}) \quad (64)$$

$$S_3 = I_+ - I_- = 0 - I_0 = -I_0, \quad \text{DOCP} = \frac{I_+ - I_-}{I_+ + I_-} = \frac{-I_0}{I_0} = -1 \quad \checkmark \quad (65)$$

For the LCP beam produced at $\chi = -45^\circ$ all signs are reversed: $I_+ = I_0$, $I_- = 0$, $\text{DOCP} = +1$.

3.6 Complete Stokes Polarimetry Protocol: General and Circular Case

Meas.	Configuration	General result	Stokes relation	RCP value
I_0	No optics	S_0	$S_0 = I_0$	I_0
I_H	LP at $\theta = 0^\circ$	$\frac{1}{2}(S_0 + S_1)$	$S_1 = 2I_H - S_0$	$I_0/2$
I_V	LP at $\theta = 90^\circ$	$\frac{1}{2}(S_0 - S_1)$	$S_0 = I_H + I_V$	$I_0/2$
I_{45}	LP at $\theta = 45^\circ$	$\frac{1}{2}(S_0 + S_2)$	$S_2 = 2I_{45} - S_0$	$I_0/2$
I_{135}	LP at $\theta = 135^\circ$	$\frac{1}{2}(S_0 - S_2)$	check: $I_{45} + I_{135} = S_0$	$I_0/2$
I_+	QWP(+45°)+LP(0°)	$\frac{1}{2}(S_0 + S_3)$	$S_3 = I_+ - I_-$	0
I_-	QWP(-45°)+LP(0°)	$\frac{1}{2}(S_0 - S_3)$	DOCP = $(I_+ - I_-)/(I_+ + I_-)$	I_0

Table 5: Complete Stokes polarimetry protocol. The general result column applies to any incident polarization state. The RCP value column gives the numerical outcome for the RCP beam $\mathbf{S} = I_0(1, 0, 0, -1)^T$ produced by the crystal QWP at $\chi = +45^\circ$. θ is the polarizer transmission-axis angle measured from horizontal.

This six-measurement (or seven, counting I_0) protocol is the standard **Stokes polarimetry** procedure used at X-ray synchrotron beamlines. In practice S_0 need not be measured separately since $S_0 = I_H + I_V = I_{45} + I_{135} = I_+ + I_-$, providing three independent consistency checks.

4 Crystal Phase Plate with General Amplitudes r_s, r_p

4.1 Physical Context

In the X-ray regime, a birefringent crystal phase plate exploits the difference in reflectivity (and phase shift) between the σ (s) and π (p) polarization components near a Bragg reflection [3, 14]. The two eigenpolarizations of the crystal are:

- **σ -polarization** (s): electric field perpendicular to the diffraction plane — mapped to the x -axis.
- **π -polarization** (p): electric field in the diffraction plane — mapped to the y -axis.

The complex reflection (or transmission) amplitudes r_s and r_p are generally different in both magnitude and phase:

$$r_s = |r_s| e^{i\phi_s}, \quad r_p = |r_p| e^{i\phi_p} \quad (66)$$

The key quantity is the **relative phase retardation**:

$$\Delta\phi = \phi_p - \phi_s = \arg(r_p) - \arg(r_s) \quad (67)$$

With this definition, the cross-product that appears throughout the Mueller matrices below satisfies

$$\text{Re}(r_s r_p^*) = |r_s| |r_p| \cos \Delta\phi, \quad \text{Im}(r_s r_p^*) = |r_s| |r_p| \sin(\phi_s - \phi_p) = -|r_s| |r_p| \sin \Delta\phi \quad (68)$$

so that the sign of $\Delta\phi$ directly sets the sign of $\text{Im}(r_s r_p^*)$, and hence the handedness of the circular polarization generated (Eq. 89). The same formalism applies verbatim to the transmitted (forward-diffracted) beam by replacing the reflection amplitudes r_s, r_p with the transmission amplitudes t_s, t_p .

For a perfect quarter-wave plate, $r_s r_p^* = r_p r_s^*$ and $\arg(r_p/r_s) = \pm\pi/2$. All Mueller matrix entries and Stokes output formulas below are expressed directly in terms of the complex amplitudes r_s, r_p and their conjugates, using the standard products $r_s r_s^* = |r_s|^2$, $r_p r_p^* = |r_p|^2$, and $r_s r_p^* = \text{Re}(r_s r_p^*) + i \text{Im}(r_s r_p^*)$.

4.2 Jones Matrix of the Crystal

With the crystal axes aligned along $\hat{x}$ (σ) and $\hat{y}$ (π), the Jones matrix is diagonal:

$$\mathbf{J}_0 = \begin{pmatrix} r_s & 0 \\ 0 & r_p \end{pmatrix} \quad (69)$$

Using the Pauli basis (Eq. 8), this decomposes as:

$$\mathbf{J}_0 = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} \boldsymbol{\sigma}_1 \quad (70)$$

This is physically transparent: the crystal acts as an isotropic attenuator (the $\boldsymbol{\sigma}_0$ term, proportional to the mean amplitude) plus a linear H/V birefringence (the $\boldsymbol{\sigma}_1$ term, proportional to the amplitude difference).

4.3 Crystal Oriented at Arbitrary Angle χ

Applying the rotation sandwich $\mathbf{J}^{(\chi)} = \mathbf{R}(-\chi) \mathbf{J}_0 \mathbf{R}(\chi)$ (Eq. 16) and using the SU(2) identity $\mathbf{R}(\chi) \boldsymbol{\sigma}_1 \mathbf{R}(-\chi) = \cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2$, the rotated Pauli decomposition is:

$$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2) \quad (71)$$

Rotation by χ simply rotates the birefringence axis from $\boldsymbol{\sigma}_1$ (H/V) toward $\boldsymbol{\sigma}_2$ ($\pm 45^\circ$) by the angle 2χ in Pauli/Stokes space. Expanded in matrix form:

$$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \mathbf{I} + \frac{r_s - r_p}{2} \begin{pmatrix} \cos 2\chi & \sin 2\chi \\ \sin 2\chi & -\cos 2\chi \end{pmatrix} \quad (72)$$

4.4 Mueller Matrix of the Crystal at Arbitrary Angle χ

4.4.1 Aligned Mueller Matrix ($\chi = 0$)

Applying the Pauli trace formula (Eq. 14) to $\mathbf{J}_0$:

$$\mathbf{M}_0 = \begin{pmatrix} \frac{r_s r_s^* + r_p r_p^*}{2} & \frac{r_s r_s^* - r_p r_p^*}{2} & 0 & 0 \\ \frac{r_s r_s^* - r_p r_p^*}{2} & \frac{r_s r_s^* + r_p r_p^*}{2} & 0 & 0 \\ 0 & 0 & \text{Re}(r_s r_p^*) & \text{Im}(r_s r_p^*) \\ 0 & 0 & -\text{Im}(r_s r_p^*) & \text{Re}(r_s r_p^*) \end{pmatrix} \quad (73)$$

The block structure is transparent: the upper-left block is driven by $r_s r_s^*$ and $r_p r_p^*$ (the individual intensity reflectivities), while the lower-right 2×2 block is the real 2×2 rotation matrix $\text{Re}(r_s r_p^*) \mathbf{I} + \text{Im}(r_s r_p^*) \boldsymbol{\epsilon}$ (where $\boldsymbol{\epsilon}$ is the 2×2 antisymmetric matrix), encoding the complex cross-term $r_s r_p^*$ between the two eigenmodes.

4.4.2 Rotated Mueller Matrix

Using the Mueller rotation matrix $\mathbf{R}_M(\chi)$ (Eq. 22):

$$\mathbf{M}^{(\chi)} = \mathbf{R}_M(-\chi) \cdot \mathbf{M}_0 \cdot \mathbf{R}_M(\chi) \quad (74)$$

Unlike the Jones case, the rotation couples all four Stokes components. Carrying out the full matrix product with $c_2 \equiv \cos 2\chi$, $s_2 \equiv \sin 2\chi$:

$$\mathbf{M}^{(\chi)} = \begin{pmatrix} \frac{r_s r_s^* + r_p r_p^*}{2} & \frac{r_s r_s^* - r_p r_p^*}{2} c_2 & \frac{r_s r_s^* - r_p r_p^*}{2} s_2 & 0 \\ \frac{r_s r_s^* - r_p r_p^*}{2} c_2 & \frac{r_s r_s^* + r_p r_p^*}{2} c_2^2 + \text{Re}(r_s r_p^*) s_2^2 & \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 & -\text{Im}(r_s r_p^*) s_2 \\ \frac{r_s r_s^* - r_p r_p^*}{2} s_2 & \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 & \frac{r_s r_s^* + r_p r_p^*}{2} s_2^2 + \text{Re}(r_s r_p^*) c_2^2 & \text{Im}(r_s r_p^*) c_2 \\ 0 & \text{Im}(r_s r_p^*) s_2 & -\text{Im}(r_s r_p^*) c_2 & \text{Re}(r_s r_p^*) \end{pmatrix} \quad (75)$$

Key structural features:

- $M_{00} = (r_s r_s^* + r_p r_p^*)/2$ and $M_{33} = \text{Re}(r_s r_p^*)$ are independent of χ .
- The last row $(0, \text{Im}(r_s r_p^*) s_2, -\text{Im}(r_s r_p^*) c_2, \text{Re}(r_s r_p^*))$ shows that S_3^{out} is driven by S_1^{in} and S_2^{in} (via $\text{Im}(r_s r_p^*)$) and by S_3^{in} (via $\text{Re}(r_s r_p^*)$).
- When $r_s r_s^* = r_p r_p^*$ (equal reflectivities) the upper-left block becomes $(r_s r_s^*/2 + r_p r_p^*/2) \mathbf{I}_{2 \times 2}$, and the off-diagonal entries in the first two rows/columns vanish.

4.5 Output Jones Vector for a General Incident Beam

Let the incident beam be an arbitrary Jones vector:

$$\mathbf{E}_{\text{in}} = \begin{pmatrix} E_x^{(0)} \\ E_y^{(0)} \end{pmatrix} \quad (76)$$

Applying the crystal Jones matrix (Eq. 72):

$$\mathbf{E}_{\text{out}} = \frac{r_s + r_p}{2} \begin{pmatrix} E_x^{(0)} \\ E_y^{(0)} \end{pmatrix} + \frac{r_s - r_p}{2} \begin{pmatrix} \cos 2\chi E_x^{(0)} + \sin 2\chi E_y^{(0)} \\ \sin 2\chi E_x^{(0)} - \cos 2\chi E_y^{(0)} \end{pmatrix} \quad (77)$$

The first term is an isotropic multiplication by the mean amplitude $(r_s + r_p)/2$; the second introduces differential retardation along the rotated eigenmodes. Equivalently, using Eq. (71) the output can be written in the Pauli basis as:

$$\mathbf{E}_{\text{out}} = \left[\frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2) \right] \mathbf{E}_{\text{in}} \quad (78)$$

Particular case: horizontal linear polarization. Setting $E_x^{(0)} = E_0$, $E_y^{(0)} = 0$ in Eq. (77):

$$\mathbf{E}_{\text{out}}|_H = \frac{E_0}{2} \begin{pmatrix} (r_s + r_p) + (r_s - r_p) \cos 2\chi \\ (r_s - r_p) \sin 2\chi \end{pmatrix} = E_0 \begin{pmatrix} r_s \cos^2 \chi + r_p \sin^2 \chi \\ (r_s - r_p) \sin \chi \cos \chi \end{pmatrix} \quad (79)$$

Particular case: vertical linear polarization. Setting $E_x^{(0)} = 0$, $E_y^{(0)} = E_0$:

$$\mathbf{E}_{\text{out}}|_V = \frac{E_0}{2} \begin{pmatrix} (r_s - r_p) \sin 2\chi \\ (r_s + r_p) - (r_s - r_p) \cos 2\chi \end{pmatrix} = E_0 \begin{pmatrix} (r_s - r_p) \sin \chi \cos \chi \\ r_s \sin^2 \chi + r_p \cos^2 \chi \end{pmatrix} \quad (80)$$

4.6 Output Stokes Vector for a General Incident Beam

Let the incident beam have an arbitrary Stokes vector $\mathbf{S}_{\text{in}} = (S_0^{\text{in}}, S_1^{\text{in}}, S_2^{\text{in}}, S_3^{\text{in}})^T$. Applying the rotated Mueller matrix $\mathbf{M}^{(\chi)}$ (Eq. 75):

$$\mathbf{S}_{\text{out}} = \mathbf{M}^{(\chi)} \mathbf{S}_{\text{in}} \quad (81)$$

Computing each component from Eq. (75), with $c_2 = \cos 2\chi$, $s_2 = \sin 2\chi$:

$$S_0^{\text{out}} = \frac{r_s r_s^* + r_p r_p^*}{2} S_0^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} c_2 S_1^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} s_2 S_2^{\text{in}} \quad (82)$$

$$S_1^{\text{out}} = \frac{r_s r_s^* - r_p r_p^*}{2} c_2 S_0^{\text{in}} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} c_2^2 + \text{Re}(r_s r_p^*) s_2^2 \right) S_1^{\text{in}} \\ + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 S_2^{\text{in}} - \text{Im}(r_s r_p^*) s_2 S_3^{\text{in}} \quad (83)$$

$$S_2^{\text{out}} = \frac{r_s r_s^* - r_p r_p^*}{2} s_2 S_0^{\text{in}} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) s_2 c_2 S_1^{\text{in}} \\ + \left(\frac{r_s r_s^* + r_p r_p^*}{2} s_2^2 + \text{Re}(r_s r_p^*) c_2^2 \right) S_2^{\text{in}} + \text{Im}(r_s r_p^*) c_2 S_3^{\text{in}} \quad (84)$$

$$S_3^{\text{out}} = \text{Im}(r_s r_p^*) s_2 S_1^{\text{in}} - \text{Im}(r_s r_p^*) c_2 S_2^{\text{in}} + \text{Re}(r_s r_p^*) S_3^{\text{in}} \quad (85)$$

Key observations:

- S_3^{out} is driven by $\text{Im}(r_s r_p^*)$ (coupling from S_1^{in} and S_2^{in}) and by $\text{Re}(r_s r_p^*)$ (preservation of S_3^{in}).
- **H/V linear content (S_1^{in}) generates circular polarization** via $\text{Im}(r_s r_p^*) \sin 2\chi$, provided $\chi \neq 0^\circ, 90^\circ$ and $\text{Im}(r_s r_p^*) \neq 0$.
- S_0^{out} is independent of S_3^{in} .

Particular case: horizontal linear polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 1, 0, 0)^T$. Substituting into Eqs. (82)–(85):

$$S_0^{\text{out}} = \frac{I_0}{2}(r_s r_s^* + r_p r_p^* + (r_s r_s^* - r_p r_p^*) \cos 2\chi) = I_0(r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi) \quad (86)$$

$$S_1^{\text{out}} = \frac{I_0}{2}(r_s r_s^* - r_p r_p^*) \cos 2\chi + I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} \cos^2 2\chi + \text{Re}(r_s r_p^*) \sin^2 2\chi \right) \quad (87)$$

$$S_2^{\text{out}} = I_0 \sin 2\chi \left[\frac{r_s r_s^* - r_p r_p^*}{2} + \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) \cos 2\chi \right] \quad (88)$$

$$S_3^{\text{out}} = I_0 \text{Im}(r_s r_p^*) \sin 2\chi \quad (89)$$

S_3^{out} (Eq. 89) is non-zero and comes from $M_{31} = \text{Im}(r_s r_p^*) \sin 2\chi$ acting on $S_1^{\text{in}} = I_0$. The transmitted intensity is $S_0^{\text{out}} = I_0(r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi)$ (Eq. 86). For a QWP ($r_s r_s^* = r_p r_p^* = r^2$, $\text{Re}(r_s r_p^*) = 0$) at $\chi = 45^\circ$, Eqs. (86)–(89) give:

$$\mathbf{S}_{\text{out}} = I_0 r^2 \begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \quad (\text{pure circular, } |P_C| = 1) \quad \checkmark \quad (90)$$

For the sign of $\text{Im}(r_s r_p^*)$: with $\Delta\phi = \phi_p - \phi_s = +\pi/2$ (i.e. $r_p = i r_s$), $\text{Im}(r_s r_p^*) = -r^2$, so $S_3^{\text{out}} = -I_0 r^2$ at $\chi = 45^\circ$: right circular polarization, in exact agreement with the ideal QWP of Section 2.8.

For clarity, the three-step rotation-sandwich makes the mechanism explicit. Step 1: $\mathbf{R}_M(\chi)\mathbf{S}_{\text{in}} = I_0(1, \cos 2\chi, -\sin 2\chi, 0)^T$. The $S_2^{\text{crystal}} = -I_0 \sin 2\chi$ component feeds into the $M_0^{32} = -\text{Im}(r_s r_p^*)$ entry in step 2, yielding $S_3^{\text{crystal}} = I_0 \text{Im}(r_s r_p^*) \sin 2\chi$, which is invariant in step 3.

Particular case: $+45^\circ$ linear polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 0, 1, 0)^T$:

$$S_0^{\text{out}} = \frac{I_0}{2}(r_s r_s^* + r_p r_p^* + (r_s r_s^* - r_p r_p^*) \sin 2\chi) \quad (91)$$

$$S_1^{\text{out}} = I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} - \text{Re}(r_s r_p^*) \right) \sin 2\chi \cos 2\chi \quad (92)$$

$$S_2^{\text{out}} = I_0 \left(\frac{r_s r_s^* + r_p r_p^*}{2} \sin^2 2\chi + \text{Re}(r_s r_p^*) \cos^2 2\chi \right) \quad (93)$$

$$S_3^{\text{out}} = -I_0 \text{Im}(r_s r_p^*) \cos 2\chi \quad (94)$$

For a QWP at $\chi = 0^\circ$ ($\text{Re}(r_s r_p^*) = 0$, $r_s r_s^* = r_p r_p^* = r^2$): $S_3^{\text{out}} = -I_0 \text{Im}(r_s r_p^*)$. At $\chi = 45^\circ$: $S_3^{\text{out}} = 0$.

Particular case: left circular polarization input. $\mathbf{S}_{\text{in}} = I_0(1, 0, 0, 1)^T$ (LCP):

$$S_0^{\text{out}} = \frac{I_0}{2}(r_s r_s^* + r_p r_p^*) \quad (95)$$

$$S_1^{\text{out}} = -I_0 \text{Im}(r_s r_p^*) \sin 2\chi \quad (96)$$

$$S_2^{\text{out}} = I_0 \text{Im}(r_s r_p^*) \cos 2\chi \quad (97)$$

$$S_3^{\text{out}} = I_0 \text{Re}(r_s r_p^*) \quad (98)$$

For a QWP ($\text{Re}(r_s r_p^*) = 0$): $S_3^{\text{out}} = 0$, confirming conversion to linear polarization for any χ .

These results are fully consistent with the output Jones vector of Eq. (77), as verified analytically and numerically in the companion script.

4.7 Key Result: Degree of Circular Polarization for a General Input

Using Eqs. (85) and (82), the degree of circular polarization for an arbitrary incident beam is:

$$P_C(\chi, \mathbf{S}_{\text{in}}) = \frac{\text{Im}(r_s r_p^*) (s_2 S_1^{\text{in}} - c_2 S_2^{\text{in}}) + \text{Re}(r_s r_p^*) S_3^{\text{in}}}{\frac{r_s r_s^* + r_p r_p^*}{2} S_0^{\text{in}} + \frac{r_s r_s^* - r_p r_p^*}{2} (c_2 S_1^{\text{in}} + s_2 S_2^{\text{in}})} \quad (99)$$

with $c_2 = \cos 2\chi$, $s_2 = \sin 2\chi$. Key observations:

1. $\text{Im}(r_s r_p^*)$ **drives the generation of circular polarization** from S_1^{in} and S_2^{in} .
2. $\text{Re}(r_s r_p^*)$ **preserves existing circular polarization** S_3^{in} — it equals zero for an ideal QWP.
3. **H/V linear content** (S_1^{in}) **generates circular polarization** via $\text{Im}(r_s r_p^*) \sin 2\chi$, provided $\chi \neq 0^\circ, 90^\circ$.
4. S_0^{in} **does not contribute to** S_3^{out} .

Particular case: horizontal linear polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 1, 0, 0)^T$, using Eqs. (86) and (89):

$$P_C(\chi)|_H = \frac{\text{Im}(r_s r_p^*) \sin 2\chi}{r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi} \quad (100)$$

This corresponds to the equation in Giles' thesis and in [5], written there as

$$P_C(\chi)|_H = -\frac{a \sin(\phi_s - \phi_p) \sin 2\chi}{\cos^2 \chi + a^2 \sin^2 \chi}$$

where $a = |r_p|/|r_s|$. Using $\text{Im}(r_s r_p^*) = |r_s||r_p| \sin(\phi_s - \phi_p)$, Giles' expression equals $-P_C(\chi)|_H$ of Eq. (100): the overall sign difference reflects Giles' opposite handedness convention for the circular polarization rate, and has no physical consequence.

For a QWP ($r_s r_s^* = r_p r_p^* = r^2$, $r_p = i r_s$ so $\text{Im}(r_s r_p^*) = -r^2$) at $\chi = 45^\circ$: $P_C = -1$ (RCP), in agreement with Section 2.8. ✓

Particular case: $+45^\circ$ linear polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 0, 1, 0)^T$:

$$P_C(\chi)|_{+45} = \frac{-\text{Im}(r_s r_p^*) \cos 2\chi}{\frac{r_s r_s^* + r_p r_p^*}{2} + \frac{r_s r_s^* - r_p r_p^*}{2} \sin 2\chi} \quad (101)$$

For equal reflectivities ($r_s r_s^* = r_p r_p^* = r^2$): $P_C = -\text{Im}(r_s r_p^*) \cos 2\chi / r^2$.

Particular case: left circular polarization input. For $\mathbf{S}_{\text{in}} = I_0(1, 0, 0, 1)^T$ (LCP):

$$P_C(\chi)|_{\text{LCP}} = \frac{2 \text{Re}(r_s r_p^*)}{r_s r_s^* + r_p r_p^*} \quad (102)$$

For a QWP ($\text{Re}(r_s r_p^*) = 0$): $P_C = 0$ — confirming conversion to linear polarization for any χ .

Input $\mathbf{S}_{\text{in}}$		General P_C (any r_s, r_p)	Ideal QWP ($r_s r_s^* = r_p r_p^* = r^2$, $\text{Re}(r_s r_p^*) = 0$)
H	linear	$\frac{\text{Im}(r_s r_p^*) \sin 2\chi}{r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi}$	$ P_C = 1$ at $\chi = 45^\circ$
$(1, 1, 0, 0)^T$			
$+45^\circ$	linear	$\frac{-\text{Im}(r_s r_p^*) \cos 2\chi}{\frac{r_s r_s^* + r_p r_p^*}{2} + \frac{r_s r_s^* - r_p r_p^*}{2} \sin 2\chi}$	$P_C = -\frac{\text{Im}(r_s r_p^*)}{r^2} \cos 2\chi$
$(1, 0, 1, 0)^T$			
LCP		$\frac{2 \text{Re}(r_s r_p^*)}{r_s r_s^* + r_p r_p^*}$	$P_C = 0, \forall \chi$
$(1, 0, 0, 1)^T$			
General		see Eq. (99)	
$(S_0, S_1, S_2, S_3)^T$			

Table 6: Degree of circular polarization $P_C = S_3^{\text{out}}/S_0^{\text{out}}$ in terms of the complex amplitudes r_s and r_p . The cross-product $r_s r_p^*$ fully encodes the retardation: $\text{Im}(r_s r_p^*)$ drives conversion between linear and circular polarization, while $\text{Re}(r_s r_p^*)$ governs preservation of existing circular polarization.

4.8 Conditions for Perfect Circular Polarization

For a horizontal linear input (Eq. 100), $|P_C| = 1$ requires $S_1^{\text{out}} = S_2^{\text{out}} = 0$ simultaneously (Eqs. 87–88), which leads to two independent conditions:

- Phase condition:** $\text{Re}(r_s r_p^*) = 0$, i.e. $r_s r_p^*$ is purely imaginary (crystal acts as a quarter-wave retarder).
- Amplitude-angle condition:** $r_s r_s^* \cos \chi = r_p r_p^* \sin \chi$, i.e. $\tan \chi = \sqrt{r_s r_s^* / r_p r_p^*}$.

For equal reflectivities $|r_s| = |r_p|$, condition 2 reduces to $\chi = 45^\circ$. For unequal reflectivities, the crystal must be tilted to compensate.

For a $+45^\circ$ linear input, condition 1 alone ($\Delta\phi = \pm\pi/2$) is sufficient for $|P_C| = 1$ at $\chi = 0^\circ$. For a general input $(S_0, S_1, S_2, S_3)^T$, perfect circular polarization is achieved whenever the numerator of Eq. (99) equals $\pm S_0^{\text{out}}$.

Input	χ	$\Delta\phi$	$ r_s $ vs $ r_p $	P_C	Output
H linear	0° or 90°	any	any	0	Linear (no conversion)
	45°	$\pm\pi/2$	$ r_s = r_p $	∓ 1	Pure RCP/LCP
	45°	$\pm\pi/2$	$ r_s \neq r_p $	$ P_C < 1$	Elliptical
	45°	0 or π	any	0	Linear
	$\arctan(r_s / r_p)$	$\pm\pi/2$	any	∓ 1	Pure RCP/LCP
$+45^\circ$ linear	0° or 90°	$\pm\pi/2$	$ r_s = r_p $	± 1	Pure LCP/RCP
$+45^\circ$ linear	any	0	any	0	Linear (unchanged)
LCP	any	0	any	+1	LCP (unchanged)
LCP	any	$\pi/2$	$ r_s = r_p $	0	Linear (orientation set by χ)

Table 7: Output polarization state for selected input polarizations, crystal orientations and parameters. The H linear results use Eq. (100); others use the general Eq. (99).

4.9 Summary

Stage / case	Jones vector E	Pauli of J	decomp.	Stokes vector / Mueller matrix
Input				
(horiz. linear)	$E_0 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$	—		$\mathbf{S}_{\text{in}} = I_0 \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$
Ideal QWP				
$(\chi = 45^\circ,$				
$ r_s = r_p = r,$				
$\Delta\phi = \pi/2)$	$\frac{E_0 r}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$	$\mathbf{J}_{\text{QWP}}^{(45^\circ)}$	$= \mathbf{S}_{\text{out}} = I_0 r^2$	$\begin{pmatrix} 1 \\ 0 \\ 0 \\ -1 \end{pmatrix} \text{ (RCP)}$
		$\frac{r(1+i)}{2} \boldsymbol{\sigma}_0$		
		$+ \frac{r(1-i)}{2} \boldsymbol{\sigma}_2$		
				$\mathbf{M}_{\text{QWP}}^{(45^\circ)} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$

Table 8: Unified summary. $\Delta\phi = \phi_p - \phi_s$ is the relative phase retardation between the π and σ eigenmodes.

Stage / case	Jones vector E	Pauli of J	decomp.	Stokes vector / Mueller matrix
Crystal @ $\chi = 0$				
$(r_s, r_p \text{ general})$	$E_0 \begin{pmatrix} r_s \\ 0 \end{pmatrix}$	$\mathbf{J}_0 = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_s - r_p}{2} \boldsymbol{\sigma}_1$	$\mathbf{M}_0 = \begin{pmatrix} A & B & 0 & 0 \\ B & A & 0 & 0 \\ 0 & 0 & C & D \\ 0 & 0 & -D & C \end{pmatrix}$	$P_C = 0$ (no circular polarization)
Crystal @ arb. χ ,				
$r_s, r_p \text{ general}$	$E_0 \begin{pmatrix} r_s \cos^2 \chi + r_p \sin^2 \chi \\ (r_s - r_p) \sin \chi \cos \chi \end{pmatrix}$	$\mathbf{J}^{(\chi)} = \frac{r_s + r_p}{2} \boldsymbol{\sigma}_0 + \frac{r_p}{2} (\cos 2\chi \boldsymbol{\sigma}_1 + \sin 2\chi \boldsymbol{\sigma}_2)$	$\mathbf{M}^{(\chi)}$: see Eq. (75)	$S_0 = I_0(r_s r_s^* \cos^2 \chi + r_p r_p^* \sin^2 \chi)$ $S_3 = I_0 \text{Im}(r_s r_p^*) \sin 2\chi$ P_C : see Eq. (100)

Table 9: Continuation of Table 8

5 Experimental benchmarking

We consider a Si 111 plate of $10\mu\text{m}$ thickness.

The incident beam is linear polarized in the horizontal plane $\mathbf{S}_{\text{in}} = (1, 1, 0, 0)^T$. This is an approximation of the beam emitted by a plane undulator.

The crystal plate is set to $\chi = 45^\circ$. This allows to split the total intensity initially in the H component, into two axis at $\pm 45^\circ$ each one receiving half intensity.

The scan is in the deviation angle from Bragg position. The r_s and r_p calculated correspond to the Bragg geometry, i.e. the angle between the Bragg planes and the crystal surface (zero as it is symmetric) is smaller than the Bragg angle. We consider the transmitted beam (also called forward diffracted).

Experimental data recorded at DESY P9 beamline is in Fig. 5. P3 is not obtained experimentally, but calculated: $P3 = \sqrt{1 - P1^2 - P2^2}$. Note that the sign of the square root is arbitrary (it cannot distinguish between left or right polarization). We selected the positive one.

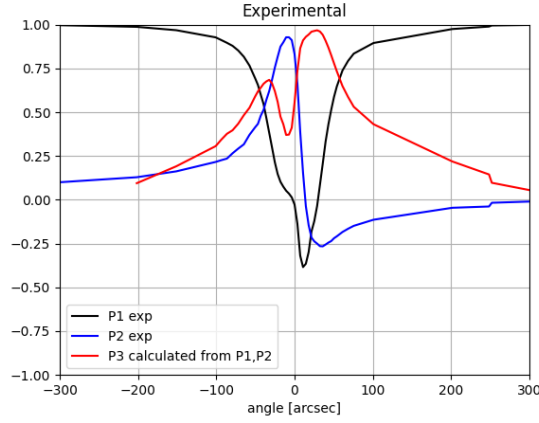


Figure 2: Experimental results.

Calculations using Eq. 75 are shown in figure 5. The reflectivities r_s and r_p are calculated using the `crystalpy` Python library [15]. We display P1, P2 and P3 (the Stokes parameters normalized to S0); P3 is the circular polarization degree. For direct comparison, the deviation angle in the calculations has been reversed, to account for the opposite sign convention of the rocking angle between `crystalpy` and the beamline scan.

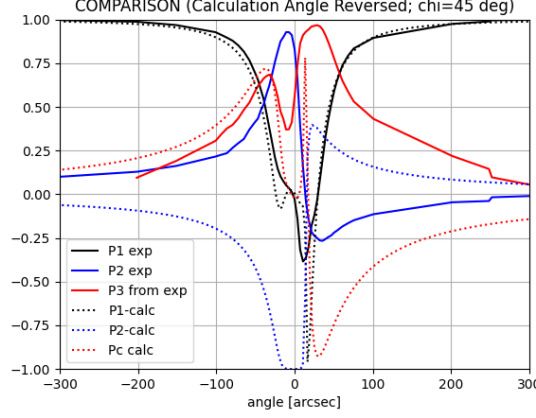


Figure 3: Calculations for $\chi = \pi/4$ and experimental results.

We can observe good agreement in P1, but in P2 and P3 the sign is changed.

For $\chi = -\pi/4$, the results are in figure 5, showing now a good agreement in both P1 and P2. The remaining sign difference in P3 is not significant: the experimental P3 is reconstructed as $\pm\sqrt{1 - P1^2 - P2^2}$ with an arbitrarily chosen positive sign, so only $|P3|$ is experimentally determined — and $|P3|$ agrees well.

These observations have a simple interpretation in terms of the rotation conventions of Section 2.4. Under $\chi \rightarrow -\chi$, the calculated P1 is even while P2 and P3 are both odd (Eqs. 87–89, all χ -dependence entering through $\cos 2\chi$ and $\sin 2\chi$). The pattern observed above — P2 flipped at $\chi = +45^\circ$ but matching at $\chi = -45^\circ$, with P1 matching in both cases — therefore indicates that the positive sense of the phase-plate rotation χ used at the beamline is opposite to the convention of Eq. (16): the experimental configuration labelled $\chi = +45^\circ$ corresponds to $\chi = -45^\circ$ in the formalism of this paper. Once this is taken into account, calculation and experiment agree in P1, P2 and $|P3|$ across the full scan.

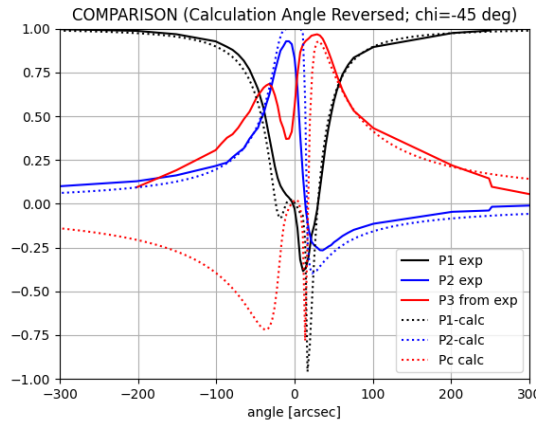


Figure 4: Calculations for $\chi = -\pi/4$ and experimental results.

Note: the calculated curves are not convoluted with the instrumental angular and energy resolution, which explains why the calculated features near the Bragg peak are sharper than the measured ones.

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